

Sprint Report and Requirement Specification

Project overview (In order of operations a user would realistically face using the app)

- 1) This project allows users to select and rent cars for a chosen period at a fixed daily rate. This is done by first asking the user to sign in with an existing account or to create a new account.
 - If the user creates a new account, they must enter their:
 - Username
 - Age
 - Password
 - Driver's license number
 - Driver's license expiry date
 - Select Membership type
 - Standard checks when creating an account
 - No fields are left blank
 - Username doesn't already exist
 - User is 21 years or older
 - Password is 8 or more characters
 - Driver's license is valid
 - Driver's license cannot already be taken
 - Membership type is selected
 - Standard Users:
 - No discounts
 - no late returns allowed (immediate penalty)
 - Late return rate of 50% per overdue day
 - Can rent only 1 vehicle at a time
 - Premium
 - 10% discount on daily rate of rentals
 - 1 day grace period on late returns
 - Late return rate of 25% per overdue day
 - Can rent 2 vehicles at a time
- 2) Once a user has created an account they can then login.
 - Standard checks when logging-in to an account
 - No fields are empty
 - Valid username and password combination exist to log user in

- Once a user is signed in, they can:
 - View vehicles available
 - Vehicles they have rented
 - Their account
 - Their cart to rent vehicles or settle penalties
- 3) If the user chooses to rent a vehicle he will be presented with the option of renting 4 types of vehicles with varying daily rates

Vehicle Type	Daily Rate
Economy	\$50
Sedan	\$70
Luxury	\$150
SUV	\$90

NOTE: Luxury vehicles can only be rented by users who are 25 years or older (there is a check in place for this when the user is viewing the vehicles).

- 4) Once the user has selected the vehicle and the number of days they want to rent the vehicle for, they can add this vehicle to the cart.
- What the cart displays
 - The name of the vehicle in the cart
 - The number of days the vehicle will be rented for
 - The total price to rent the vehicle for the duration before other charges
 - Any discounts the user might be applicable for (EX: Premium user 10% discount)
 - Any surcharges the user might be applicable for (EX: young renters' fee: <25 years old)
 - Final total after all charges are applied
 - Option to remove vehicle from cart
 - Any outstanding penalties that need to be paid
 - Checks performed on the cart
 - The cart is not empty
 - A user session exists when adding vehicles to the cart
 - Drives license has not expired
 - No more than 1 or 2 rentals at a time depending on user type.
 - Users under 25 years old cannot rent luxury vehicles
 - Vehicle isn't already rented
 - Vehicle is being rented for at least 1 day

- User has enough money in account to rent vehicles.
- 5) If the user does not have enough money in their account, they can manually add money by entering an amount. A simple check is done to ensure a negative amount isn't entered.
- 6) Once vehicles have been rented these vehicles are marked as unavailable to all other users and users can view vehicles they have rented in My Rentals
 - Information displayed in My Rentals
 - Vehicle name
 - Date vehicle was rented on
 - When rental is due to be returned
 - Is the rental still active
 - Option to return rental
- 7) Finally, users can return vehicles that have been rented from the My Rentals section.
 - Checks done on returning a vehicle
 - Vehicle hasn't already been returned
 - Vehicle is not overdue by more than 2 hours of return window
 - If overdue by more than 2 hours mark as late return
 - Premium users:
 - get 1-day late penalty grace period
 - 25% late return charge per overdue day
 - Standard users
 - No grace period
 - 50% late return charge per overdue day
 - Apply late return penalty per overdue date depending on user type
 - Send total to cart to be paid by user

User Cases

Field	Description
Use Case ID	UC-01
Use Case Name	Rent a Car
Actors	Registered Member (Standard or Premium), System

Preconditions	User must have a valid account (logged in) User has a valid driver's license User meets minimum age requirements Vehicles exist in the system with available status
Trigger	User selects a vehicle class and chooses "Rent"
Main Flow	1. User selects desired vehicle class (Economy, Sedan, SUV, Luxury) 2. System checks age eligibility and license validity. 3. System checks vehicle availability 4. If available, system assigns a return date/time 5. System calculates rental cost 6. System marks vehicle as unavailable 7. Rental record is created and stored
Alternative Flows	A1 at step 1- Age Restriction - If user is of age 21-24 Luxury vehicle is unavailable A2 at step 2 - Invalid License - if license is expired or invalid, rejects with "Driver's license not valid"
Postconditions	Success: Vehicle is reserved, user has a rental record, and cost is calculated Failure: User remains on rental page with rejection message
Business Rules / Notes	Rentals are charged in 24-hour increments Premium members receive 10% discount and can rent up to two cars at once Standard members can rent only one vehicle.

Field	Description
Use Case ID	UC-02
Use Case Name	Return a Car
Actors	Registered Member, System
Preconditions	User has an active rental
Trigger	User returns vehicle before, at or after the scheduled return time
Main Flow	1. System calculates rental duration 2. System checks for late return 3. System calculates any mileage overage charges 4. Total cost = base rental + late fees + mileage overage 5. Vehicle is marked available 6. Rental record is closed
Alternative Flows	A1 at step 2 - If late: Standard Member: Late fee = 50% of daily rate × days late Premium Member: One-day grace. After grace: 25% of daily rate × days late.
Postconditions	Success: Vehicle is returned, available for future rentals, charges applied Failure: Rental record remains open
Business Rules / Notes	Late = returned more than 2 hours past scheduled time Mileage cap: 200 miles/day Extra miles = \$0.25/mile

Field	Description
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Use Case ID	UC-03
Use Case Name	Register an Account
Actors	New User, System
Preconditions	User is not already registered
Trigger	User selects "Register" on the home page
Main Flow	1. User enters personal details (username, age, driver's license ID, password, account type i.e.. Standard, Premium) 2. System validates input format (Password length) 3. System checks that user is over 21 years old 4. System verifies driver's license isn't expired 5. System creates a new user account
Alternative Flows	A1 at step 3- if user is under 21 System rejects with message: "Must be 21 or older to register." A2 at step 4- if license is expired System rejects with message: "Driver's license invalid." A3 at step 5 - If username is already taken System rejects with "Username already taken"
Postconditions	Success: User has an active account Failure: User is not registered and must correct errors
Business Rules / Notes	Passwords must be over 8 characters Valid license is required to register

Field	Description
Use Case ID	UC-03
Use Case Name	User Login
Actors	Registered User, System
Preconditions	User already has a valid account
Trigger	User selects the “Login” option on the home page
Main Flow	1. User enters username and password 2. System checks credentials against user database 3. If valid, system creates a session 4. System redirects user to the home page
Alternative Flows	A1 - At Step 3, if credentials do not match System shows: “Invalid username or password.” User may retry
Postconditions	Success: User logged in with active session. Failure: User remains on login page with error message
Business Rules / Notes	Check for valid user name and password with an already registered user

Field	Description
Use Case ID	UC-04
Use Case Name	View Rented Vehicles
Actors	Registered Member (Standard or Premium), System
Preconditions	User must have a valid account (logged in) User must have already rented a vehicle

	Vehicles exist in the system and is marked as rented
Trigger	User selects “My Rentals” on their account
Main Flow	1. User views vehicles they have rented on “My Rentals” page 2. Page displays vehicle name, rented date, return date and status to user 3. User is given the option to return the rental
Alternative Flows	- None
Postconditions	UC-02 (Return a car) is invoked
Business Rules / Notes	- Same as UC-02

Sprint Report

Overall, this sprint was successful in covering required criteria for project milestone 1 within the given period. For this project the team implemented:

- A robust back end with business logic containing criteria specific to 2 types of users on the app
- A very simple but functional UI with safeguards to help mitigate app crashes
- 4 unique types of rentals which are controlled by age checks
- Discount logic for premium users
- Surcharges for young renters
- Calculated return windows by dates and times
- Late return grace period for premium users
- Varying late return rates based on user type
- Checking finances before processing transactions

Features and Areas covered by each team member

Hiruna Yevin Dissanayake

- Application Setup
- Login
- Registration
- User authentication and sessions
- Cart and checkout logic

Tyson Russell

- Vehicles.html page
- Vehicle constructors
- Vehicles data entry
- Rental service

Ivan Quintero

- Setting up vehicle and rental entities
- Business Logic in backend
 - o Vehicle checkout
 - o Vehicle return
 - o Calculating late fees
 - o Calculating additional fees if applicable
 - o Calculating total charge